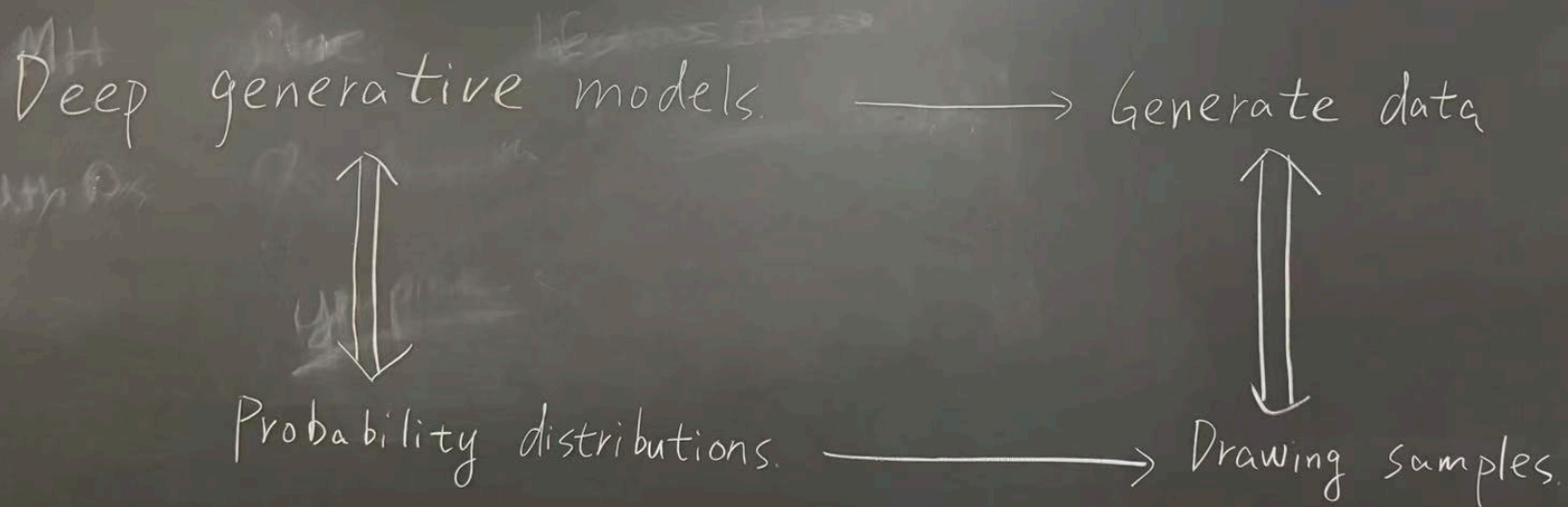




key

① De

② P

③

④

erate data



Drawing samples.

key tasks.

- ① Design a model.
- ② Parameter learning
- ③ Generate data
- ④ Understand data.

} building
the
model

} Using
the model
inference)

Task / problem

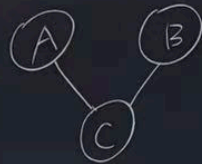
- ↓
- 1 Abstract the task into
 - Objects
 - Relations

④

Task / problem



1. Abstract the task into a graph
- Objects. → Nodes
 - Relations → Edges



2. Associate random Variables to nodes



Categorical variable

$$\begin{aligned}P(X_C = S) &= 0.1 \\P(X_C = U) &= 0.8 \\P(X_C = I) &= 0.1\end{aligned}$$

A graph of random variables



Probabilistic graphical models

Edges

Bayesian network

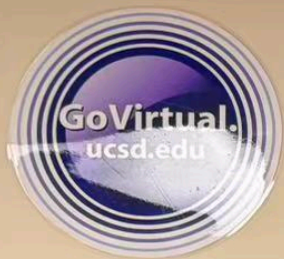
— Directed → causality



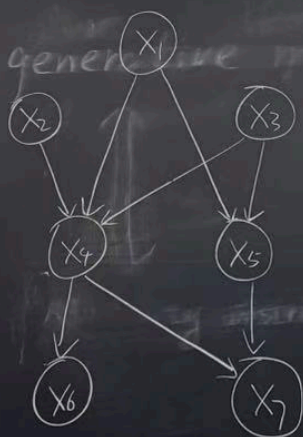
— Undirected → correlation



Markov
Random
Fields



Directed
acyclic
graph (DAG)



3. Define local distributions

$$p(x_1) \quad p(x_2) \quad p(x_3)$$

$$p(x_4 | x_1, x_2, x_3) \quad p(x_5 | x_1, x_3)$$

$$p(x_6 | x_4) \quad p(x_7 | x_4, x_5)$$

4. Joint distribution

$$p(x_{1:7}) = p(x_1) p(x_2) p(x_3) p(x_4 | x_1, x_2, x_3) p(x_5 | x_1, x_3) p(x_6 | x_4) p(x_7 | x_4, x_5)$$

I-map

Example: large language model

$p(\text{sentence})$

1 Graph

- Objects : words in the
- Relations :

Example: large language model

$P(\text{sentence})$

↑ Graph

- Objects: words in the
- Relations:

$$p(x_1, x_2, x_3) p(x_5 | x_1, x_3) p(x_6 | x_4) p(x_7 | x_4, x_5)$$

Task/problem

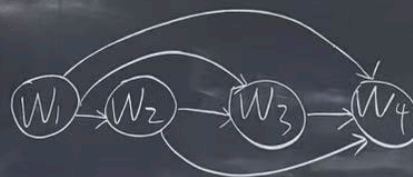


1 Abstract the task into a graph

- Objects → Nodes

(w_N) - Relations → Edges

sentence



I like apple because it is ?

handsome X
delicious ✓
Smelly X

Task / problem



1. Abstract the task into a graph

— Objects → Nodes

(w_4) ... (w_N) — Relations → Edges

because it is — ?

handsome X
delicious ✓
Smelly X

2. Associate random variables to nodes



$V = 10,000$

$P(w_4 = \text{machine}) = 0.6012$

$P(w_4 = \text{cup}) =$
dog

Categorical variables

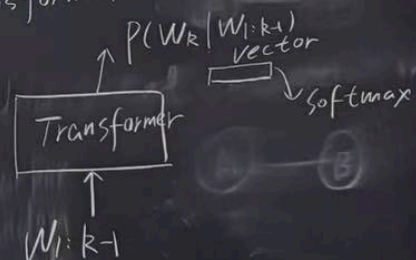
$P(w_1)$

$P(w_k | w_{1:k-1})$

$$P(w_{1:N}) = P(w_1) \prod_{k=2}^N P(w_k | w_{1:k-1})$$

Transformer

Transformer





Variational autoencoder (VAE)

Deep generative model

Task: generate images

2. Random Variable

Image

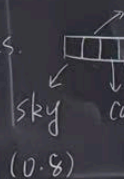
r_1
r_2
r_3
r_4
r_5

1. Graph

- Objects
- Relations

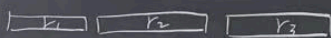
{ Image (matrix of pixels) X
Semantics (sky, building, ...) Z

Semantics



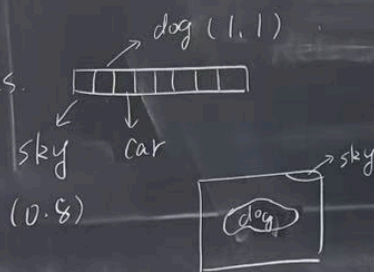
2. Random variables.

Image



Multi-variant Gaussian distribution

Semantics



3 Joint distribution

$$p(X, Z) = p(Z) p(X|Z)$$

4. Local distribution

$$p(Z) = N(Z|\vec{0}, \Sigma)$$

$$p(X|Z)$$

$$N(X|\vec{0}, \Sigma)$$

Gaussian distribution

3 Joint distribution

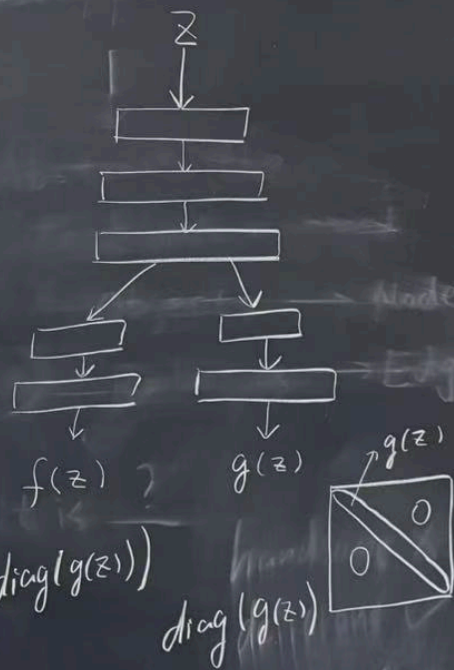
$$p(x, z) = p(z) p(x|z)$$

4 Local distributions

$$p(z) = N(z|\vec{0}, I)$$

$$p(x|z)$$

$$N(x | f(z), \text{diag}(g(z)))$$



2. Associate Variables

Gaussian X_A

$$V = 10,000$$

$$p(w_4 = \text{machine}) =$$

$$p(w_4 = \text{cup}) =$$

dog